

Forum Brings Fresh Air to Ventilation Issues

So far in the United States, only Minnesota and Washington require mechanical ventilation systems to be installed in new homes, while all new houses in Canada must have such systems. In Washington, recent evaluations show that fewer than half of the systems perform as they should—and some perform so badly that they actually make indoor air quality (IAQ) worse. Meanwhile, a new standard on mechanical ventilation systems proposed by the American Society of Heating, Refrigeration, and Air Conditioning (ASHRAE) has been hotly debated by industry and building scientists.

All this activity raises some fundamental home performance questions: Do building scientists agree on why ventilation systems should be installed? Which is the best type of

system to choose? and What are the potential problems? We asked four experts from across the nation to weigh in on these issues.

Neil Moyer is currently employed by the Florida Solar Energy Center; has co-authored numerous scientific papers and articles relating to building diagnostics and repair; and has been a trainer for many utilities, the Energy Efficient Building Association, and Affordable Comfort.

Armin Rudd is principal engineer at Building Science Corporation. His current activities are primarily in the U.S. Department of Energy's Building America Program; most of the information he presents was collected through Building America research. Armin has spent 15 years in buildings research and consulting, and before that was a residential building subcontractor.

Max Sherman is group leader of the Energy Performance of buildings group in the Indoor Environment Department of the Energy Efficiency Technologies Division at Lawrence Berkeley National Laboratory. He is also chairperson of ASHRAE's Residential Ventilation Standard, SPC 62.2P.

Don Stevens is a private consultant who works on energy and ventilation codes, products, and standards on behalf of manufacturers, utilities, and agencies. He has been active in ventilation code and product development for nearly 20 years.

Here's what they had to say.

[Editor's note: Answers were edited for conciseness and clarity. For more details on the topics raised here, send me your questions at colleen@home-energy.org, and I will try to have them addressed in future issues.]



Neil Moyer



Armin Rudd



Max Sherman



Don Stevens

STEVENS & ASSOCIATES

Tight homes need fresh air, but which homes need mechanical ventilation systems, and which system should you choose?

Home Energy: *What is the purpose of installing a whole-house mechanical ventilation system in a home?*

Don Stevens: The purpose of mechanical ventilation is to remove contaminants and introduce cleaner outdoor air—independent

of the weather, the house construction, the building site, and the occupants' actions.

Neil Moyer: This assumes that localized pollution sources have been addressed by a spot ventilation system.

Armin Rudd: Excessive moisture is a pollutant, so mechanical ventilation

should control interior moisture. Another purpose is to control interior pressure with respect to the outside, in order to maximize building durability, aid combustion safety, and improve IAQ. Finally, ventilation can be done for the sensory satisfaction of the occupants.

Max Sherman: Another reason to install a mechanical ventilation system is to remove internal thermal gains (for ventilative cooling).

HE: *What are the qualities of a good ventilation system? That is; what should a contractor strive for (and a homeowner expect from) a good ventilation system?*

Don Stevens: One that introduces a controlled amount of outdoor air quietly and with good distribution while remaining “transparent” to the occupant. By this I mean that the fans are not noisy, that drafts are minimized or eliminated, that occupant interaction is not needed to operate the system, and that the system is cost-effective.

Max Sherman: Systems that cause draft or noise will be disabled. High maintenance systems will not be maintained properly and therefore will become unusable. Systems that require much user intervention will be ignored. It is important to remember that the ventilation system is part of the whole-house system and must fit into the home’s design.

Armin Rudd: The system should have at least a 15-year useful life at the design operational time.

Neil Moyer: It should provide indication of operation (or nonoperation); allow the occupants to shut it off; filter the outside air brought into the house; be accessible and have replaceable or serviceable parts; and not cause problems in the areas of health and safety, building durability, comfort, or energy usage.

HE: *Should mechanical ventilation be standard in homes built today?*

Don Stevens: Yes, I believe that all houses built in North America should have mechanical ventilation systems both for general indoor air quality and for spot ventilation in bathrooms and kitchens.

Max Sherman: All new homes should have designed ventilation systems, whether they be mechanical or passive. Passive ventilation options can work in many situations, but they may require so much design or commissioning that it is more practical to put in a simple mechanical system.

Armin Rudd: Most people like their homes better when they are not

expensive, drafty, stuffy, odoriferous, moldy, and wet. They like their houses to be affordable, comfortable, healthy, and durable. To accomplish this, random natural infiltration should be minimized, and controlled mechanical ventilation should be employed.

HE: *Armin, do you mean no one should ever open their windows?*

Armin Rudd: No. Operable windows are useful and needed. They are needed for egress in an emergency, and they are useful for enjoying mild weather, yelling at the dog/kids/neighbors, and that occasional major airing-out of the house.

HE: *What minimum criteria would determine whether or not a new home needs mechanical ventilation? In what cases would you retrofit a home for mechanical ventilation?*

Don Stevens: Virtually all houses built today are relatively tight compared to houses built 50 years ago, even if no overt attempts have been made to make them tight. As a consequence, virtually all new houses need mechanical ventilation. I guess a minimum criterion could be that if a home has heating or cooling installed, it needs mechanical ventilation. Older homes should be retrofitted for mechanical ventilation if there are moisture problems, if there is any overt air tightening being done, or if new windows or doors have been installed.

Armin Rudd: For new construction, a minimum criterion could be demonstration of the greater of 40 ft³ per minute continuous or 10 ft³ per minute per person continuous, or the equivalent volume over a two-hour period, of unpolluted outside air. The air must be distributed uniformly throughout the house while it is occupied.

A retrofit criterion for existing construction is much harder to determine. A first criterion would be that

the homeowner be willing to pay for it. A second might be that the home tests at less than 4–5 ACH at 50 Pa pressure difference.

Max Sherman: ASHRAE Standard 62 provides minimum ventilation rates for all houses. The proposed residential version of this standard (62.2) is currently in draft form and, when issued, will provide better guidance. These standards set the rates according to the size of the home and the number of occupants. If proper ventilation rates are not met through passive means, it requires that a mechanical ventilation system be installed. Sometimes there are reasons to increase ventilation rates above these minimum requirements.

Existing houses tend to have much higher infiltration rates than new ones. Therefore, in most existing houses, it would be a waste to install a mechanical ventilation system, unless the system was addressing a specific IAQ concern. In

the absence of such concerns, the airtightness of an existing house can be measured with a blower door, and then ASHRAE Standard 136, which converts such data into ACH rates, can be used to determine if additional ventilation is needed.

Neil Moyer: New Construction: All homes should have

spot (exhausted to outdoors) ventilation in the kitchen and in bathrooms with tubs or showers. Whole-house ventilation systems should be installed in homes where long-term annual estimated air change rates are less than about 0.2 (that is, 4–5 ACH at 50 Pa). The amount of ventilation air required should be based on the number of occupants: The long-term average of the system should probably be around 7–10 CFM per person (it would probably be number of bedrooms plus 1). Retrofit: Probably similar to new construction, but now specific occupant life-style can be addressed—questions like smoking, actual occupancy load, hobbies, and so on.

All houses built in North America should have mechanical ventilation systems for both the general indoor air quality of the house and for spot ventilation needs in bathrooms and kitchens.

HE: Of the three main types of mechanical ventilation (see "Mechanical Ventilation: Three Types," p. 18), certain types work better in certain climates and markets. What type of system would you recommend for typical homes in the following locations: Phoenix, Arizona; Seattle, Washington; Fargo, North Dakota; Tampa, Florida?

Neil Moyer: Phoenix: Any. Seattle: Any. Fargo: Balanced or exhaust. Tampa: Supply or balanced.

Max Sherman: Climate is only one of many issues that factor into the selection of ventilation systems. One must design the house as a system and include the type of ventilation in that design. Climate itself affects comfort, energy costs, mold growth, and so on.

To meet the minimum requirements of the proposed Standard 62.2 or similar standards, the first system type one should look at in any climate is continuous exhaust ventilation. However, there are reasons to consider other options in some climates. For example, in a hot, humid climate like that in Tampa, supply ventilation could be a better solution, particularly if you have an envelope design that is susceptible to moisture intrusion. Similarly, in an extreme climate with higher ventilation-related energy loads such as that in Fargo, a balanced system with heat exchange could be more cost effective.

Don Stevens: Phoenix: Supply ventilation, to be integrated with the air conditioning. Seattle: Exhaust ventilation with window ports or wall vents. Fargo: Balanced ventilation with heat recovery because of the extreme climate. Tampa: Supply ventilation, in order to avoid pulling humid air into the wall assembly where it can condense and cause mold to grow. Also, consider a balanced ventilation system with an energy recovery ventilator to dehumidify the incoming outdoor air.

Armin Rudd: Phoenix, Production homes: Central-fan-integrated supply ventilation (see Figure 1). Phoenix, Custom homes: A heat recovery ventilation (HRV) unit combined with a fan recycling system on the central air handler. Seattle: Same as for Phoenix. Fargo, Production homes: A continuous-exhaust fan pulling air from the central area, combined with a lower level of central-fan-integrated supply ventilation with fan recycling. Fargo,

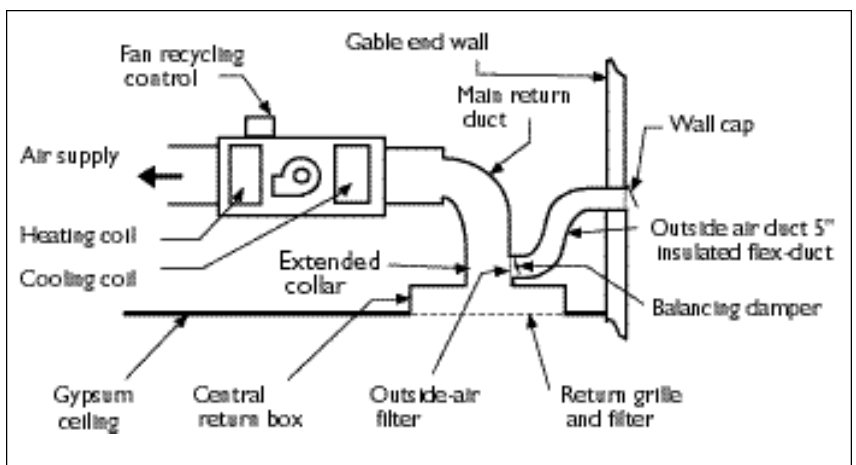


Figure 1. This central-fan-integrated supply ventilation system would be an affordable option for homes in a climate such as that in Phoenix, Arizona. It has a 5- to 8-inch insulated outside air duct connected to the return side of the central air distribution system. A balancing damper in the outside air duct field-adjusts the air flow upon system startup. The outside air is filtered before it enters the central fan.

Custom homes: An energy recovery ventilation (ERV) unit combined with a fan recycling system on the central air handler. Tampa, Production homes: Central-fan-integrated supply ventilation, with central fan recycling, and a stand-alone dehumidifier. Tampa, Custom homes: A high-efficiency dehumidifier that filters and preconditions a mixture of inside and outside air and delivers the ventilation air to the central area, with central fan recycling for whole-house circulation and mixing.

HE: Other than climate, what else must be considered in choosing a system?

Don Stevens: Cost is a consideration. Exhaust ventilation is the most familiar and generally the least expensive to install and operate. A quiet bath fan can provide both general ventilation and spot ventilation for the bathroom. However, in hot, humid climates, exhaust ventilation can pull humid outdoor air into the wall assembly, where it can condense on the back side of air-conditioned drywall and cause mold to grow, so durability issues also come into play. It also can increase the potential for backdrafting combustion appliances, causing health concerns.

Supply ventilation is useful in hot, humid climates to avoid condensation and can be integrated with a forced-air furnace or A/C air handler for

tempering, filtering, and distribution. If there is no forced air system, dedicated ductwork is necessary to ensure distribution to the habitable rooms. This increases cost.

Balanced ventilation has many positive attributes related to thermal comfort, filtering, distribution, and energy efficiency. It can be a simple power-in/power-out system using two fans, or it can include heat or "coolth" recovery. It can also be used for moisture recovery, either to dehumidify incoming hot, humid air or to transfer moisture to the incoming dry air in cold climates. However, it is generally the most expensive option.

Armin Rudd: Health is an essential consideration. Exhaust ventilation systems draw outside air from leakage openings and pathways located randomly throughout the building envelope; it is not possible to treat the outside air before it enters the living space. The ventilation air could be fresh and healthy, or it could be coming from locations with high pollutant concentrations. For example, ventilation air drawn from a garage, crawlspace, or basement sump, or from underneath a concrete slab, may induce entry of fuel vapors, combustion gases, insecticides, radon gas, excessive water vapor, and fungal or mold spores.

The source of ventilation air is known for supply and balanced sys-

tems. The air can also be heated, cooled, dehumidified, filtered, and/or cleaned before distribution to the living space. Therefore, it is easier to control the air quality factors. Regarding cost and maintenance, an advantage of central-fan-integrated ventilation is that the same air handler that distributes air for heating and cooling is used to distribute ventilation air, and that the fan and filter system are likely to be kept in working condition and to be used.

HE: Exhaust systems are the mechanical ventilation system most frequently installed in homes. How severe are the hazards associated with house depressurization caused by exhaust ventilation?

Don Stevens: Backdrafting does happen, but the magnitude of the problem has probably been overstated. The largest backdrafting potential comes from oversized exhaust fans, such as the 600–1,200 CFM range hoods used in many custom homes. Sealed-combustion appli-

ances minimize the potential for backdrafting. Using a balanced mechanical ventilation system, at least for homes with large flow devices such as range hoods, also reduces that potential. However, short chimneys, chimneys located on cold exterior walls, and multiple chimneys can all be backdrafted easily by wind forces, which can be much stronger than the 2–5 Pa that a typical exhaust system would impose on the house. A balanced ventilation system provides bal-

Mechanical Ventilation: Three Types

There are three basic types of whole-house mechanical ventilation systems: exhaust, supply, and balanced.

Exhaust: An exhaust system may consist of a single fan that is centrally located in a hallway or at the top of the stairs, or it can be an upgraded bathroom fan that has a motor built to run for longer periods of time and is sized to ventilate the whole house. If the latter, it should be a two-speed fan that can be operated continuously at low speed to provide general ventilation and switched to high speed when the room is being used. (Other bathrooms still need their own spot exhaust fans.)

Another method is installing a remote multi-point exhaust fan. Usually located in the attic, a remote fan is ducted to several exhaust grilles that take air from locations throughout the house, including each bathroom (see Figure A). This system replaces several noisy, inefficient bath fans. Passive vents in the walls provide fresh air inlet.

Supply: A supply system slightly pressurizes the house by drawing outdoor air into the house. This pressurization pushes indoor air out through small openings in the building, which helps to prevent outside moisture and pollutants from getting in through these passages. Positive indoor pressure also protects against backdrafting.

To set up a central supply system, you can mount a single fan on the wall to pull outdoor air into one room, or install a remote fan with ductwork to the most frequently used rooms—generally living rooms and bedrooms (see Figure B).

For an existing house, you can run a duct from the outside and connect it to the return duct of an existing forced-air heating and cooling system. A variable-speed integrated control motor (ICM) fan allows a reduced air flow for ventilation when heating or cooling is not needed. However, it may be best to get a fan specifically dedicated to ventilation, keeping it separate from the heating and cooling system. All of these approaches vary in terms of expense and effectiveness.

Balanced: A balanced central system uses two fans and creates a neutral, or balanced, pressure. One fan exhausts air out of the house, while the other brings the same amount of outside air into the house. They may be two wall-mounted fans of equal capacity—one for supply and one for exhaust—in two different rooms. Alternatively, they may be ducted to supply air to the common living areas, such as living rooms and bedrooms, and to exhaust air from the rooms where pollution is high, such as the kitchen and bathroom (see Figure C).

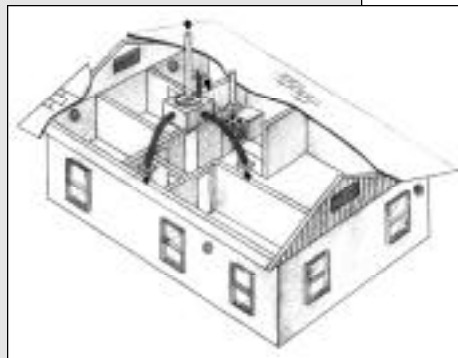


Figure A. An exhaust ventilation system typically consists of just one fan and an outlet vent but may consist of a remote fan with intakes at several locations, as shown here. Passive vents may be installed in the home to provide clean, controlled ventilation air.

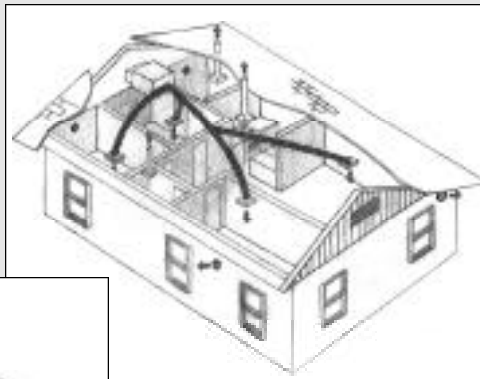


Figure B. In a supply system, a fan pulls outside air into the house, creating positive pressure. Ductwork brings the air into the living spaces that need it most—typically the bedrooms and living room. Spot exhaust fans are still needed in high-pollutant areas such as the kitchen, bathroom, and laundry room.



Figure C. In a balanced system, air is supplied to some rooms and exhausted from others. An optional heat- or energy-recovery unit transfers heat or humidity from one airstream to the other. Spot exhaust is included where necessary; here it is in the laundry room.

anced flow under only one set of temperature, pressure, and wind effects for any given house. Once a change in temperature or in pressure occurs, the pressure characteristics of the entire house change and the flow may no longer be in balance. A heat recovery ventilator in defrost mode may well place the home under negative pressure, because it is only running the exhaust side fan.

Max Sherman: Properly sized exhaust ventilation systems rarely cause any problems, but they do pose a small potential risk that does not exist with the other options. In most cases, the problem is caused by an inappropriate whole-building design rather than an inappropriate choice of ventilation system (or any one other single thing). It is a combination of exhaust ventilation and one or more susceptible components (for example, building envelope, combustion appliance, venting system).

There are many ways of avoiding these problems (for example, by using direct-vent appliances). To reduce back-drafting susceptibility, combustion appliances could be eliminated, be moved out of the pressure boundary, or be direct vented or power vented.

Neil Moyer: The hazards range from none to severe. They depend on the amount of air exhausted, the type of equipment in the house, and the climate that the house is located in. A tight house with a naturally aspirated combustion device has the potential for flame rollout, CO generation, and excessive moisture (from combustion of fuels), as well as SO_x, NO_x, and carbon (soot) emissions.

Armin Rudd: Building depressurization in hot, humid climates—other than that from normal intermittent operation of bathroom fans, kitchen range hoods, and dryers—is hazardous. Use of low-permeability interior finishes increases the problem greatly. Humid outdoor air will randomly enter the building envelope, even following circuitous paths within

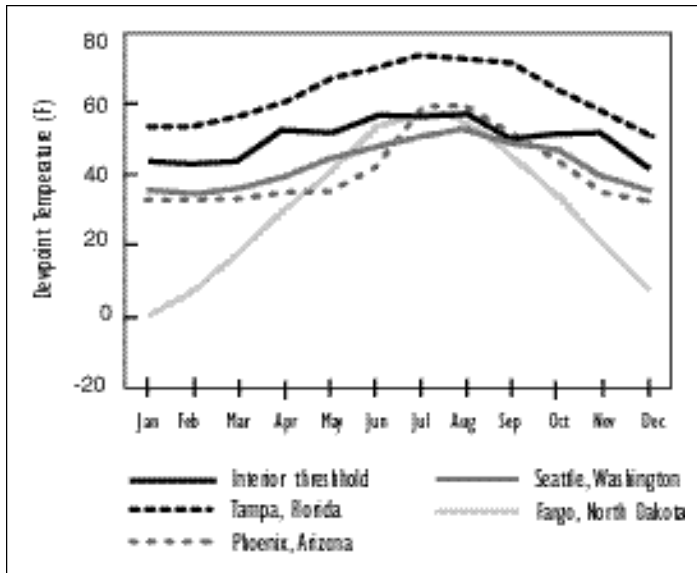


Figure 2. When choosing an energy recovery mechanical ventilation system, it is important to determine the mean monthly outdoor dew point temperature for the location, and to keep in mind the interior dew point threshold temperature.

partitions and interstitial spaces, where it comes in contact with surfaces that are below the air dew point temperature (see Figure 2). If those materials stay moist long enough, mold will result.

Combustion appliances inside houses should be at least power vented, and preferably sealed combustion. In such houses, it would be unlikely for whole-house exhaust ventilation systems alone to cause a problem with combustion safety. Whole-house exhaust ventilation combined with other exhaust devices can cause more severe depressurization, requiring an intended makeup air source. Such devices

would include a clothes dryer, a kitchen range hood or cooktop downdraft fan, a bathroom fan, a power-vented gas hot water heater, or a power-vented furnace.

For more information:

Roberson, Judy, et al. *Recommended Ventilation Strategies for Energy-Efficient Production Homes*, LBNL 40378. Berkeley, California: Lawrence Berkeley National Laboratory, December 1998.

Stevens, Don. "Mechanical Ventilation for the Home." *Home Energy* (March/April 1996):13.